



PI ZERO WORLD WIDE WEB

Streaming On Mobile Hotspot

SOMNATH BERA

Webcasting on a server-grade machine such as a laptop or desktop is routine and unremarkable. Streaming through modem port forwarding is similarly straightforward. What is far more challenging, however, is achieving stable, long-term webcasting using only a mobile phone as a Wi-Fi hotspot and a compact Raspberry Pi Zero instead of a full-sized computer. This was accomplished with a Raspberry Pi Zero equipped with just 500MB RAM and a 5MP Pi Zero camera—small in form, yet powerful in function.

This portable system delivers one-to-one streams globally without dependence on proprietary software or paid services (beyond standard mobile internet charges). Streaming is powered by the open source Motion software. The same setup can also double as a surveillance tool, capturing images and videos of intruders with date and time stamps for added security.

The Raspberry Pi Zero web camera system integrates compact hardware with open source software

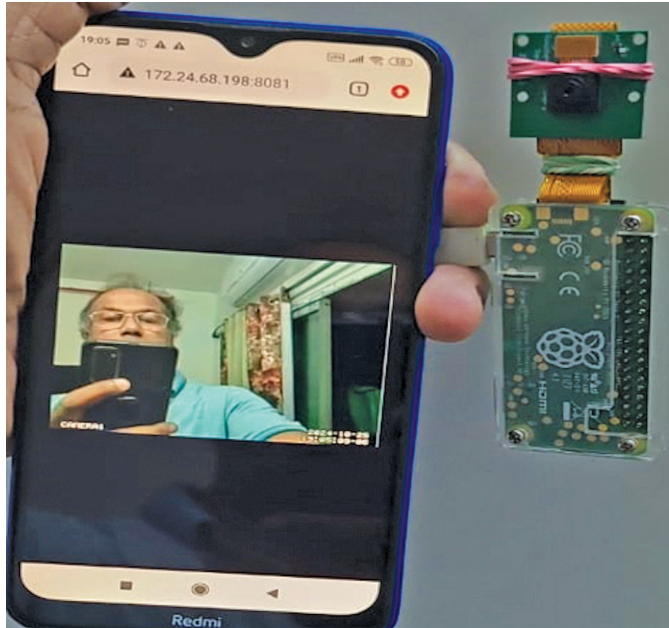


Fig. 1: Author's prototype

to provide continuous video streaming and motion detection. Fig. 1 shows the prototype. The required components are listed in the Bill of Materials table.

In addition to the components listed in Bill of Materials table, basic soldering tools, including soldering wire, a soldering iron, and soldering paste are also required.

Hardware setup

The system consists of a Raspberry Pi Zero board connected to a 5MP Pi camera module, as shown in Fig. 1. The Pi Zero accesses the internet via a mobile phone acting as a Wi-Fi hotspot. This compact and portable configuration enables the Pi Zero to function either as a web camera or a motion detection system.

Schematic circuit: A Raspberry Pi Zero connected to a camera via its camera port.

Software components: Motion and ZeroTier

Motion. The core of the system is the open source Motion software. It includes a built-in web server that handles video streaming, motion detection, and recording. Motion captures video streams from the camera and processes them in real time. When motion is detected, it can trigger actions such as recording video, taking snapshots, and saving them locally or transmitting them over the network.

Several other software solutions, including Python and OpenCV-

based setups, were evaluated. However, apart from Motion, all others proved too heavy for the Raspberry Pi Zero, resulting in unreliable streaming performance.

On the Raspberry Pi Zero, the following steps are required:

```
$ sudo apt install motion -y # Install motion
$ motion -h # Check the version of motion after installation
$ sudo systemctl start motion # Start the motion service
$ sudo systemctl enable motion # Enable motion to start automatically at boot
```

To check whether Motion is running on the local network (LAN), open a web browser on any computer connected to the same Wi-Fi network and visit: http://<your_pi_ip_address>:8081 This displays the live video stream from the Motion server.

```
$ sudo systemctl stop motion # Stop the
```

BILL OF MATERIALS

Components	Quantity
Raspberry Pi Zero W	1
5MP Pi DSI Zero camera	1
RPi camera DSI ribbon cable	1
SD card (8GB minimum, 32GB recommended)	1
2A, 5V Raspberry Pi power adaptor	1